EVERY AP-Robo question (portal, till now)

Generated: 2026-10-24T01:57:36 by `scrape.py --export AP-Robo`

For the AI tutor: teach these in due-date order. Status and XP are the student's real portal state. Question text is included in a code block wherever the tracker captured it; questions without text are listed at the end.

Applied Physics 1 - Contest 2 (Subjective) [Contest] — due 2026-10-16 11:20

Centre of Mass — 0/40 XP — NOT SOLVED — deadline 2026-10-16 11:20

URL: https://my.newtonschool.co/playground/subjective/9gkqeb7ttnk6

NSTB'26-CS

Total XP

5,795

Leaderboard Unlocked

Rank Increased

Back

CENTRE OF MASS

A cubical block of ice of mass (m) and edge length (L) is placed on a large horizontal tray of mass (M). The ice melts completely, and the resulting water spreads uniformly into a negligibly thin layer over the tray's base.

Without performing any numerical calculations, explain whether the centre of mass of the combined system (ice + tray) changes after melting. In your answer, discuss how the redistribution of mass affects the centre of mass and how the relative masses of the ice and tray influence the magnitude of this change.

(Instructions: Justify your answer using physical reasoning. No mathematical derivation is required.)

Submit

AI CODELENS

BETA

Let’s see how you answered

Correctness

Your answer should be free from errors and misinformation.

1/5

Completeness

Your answer should contain all the key points.

1/5

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Our feedback and suggestions

The answer claims that the centre of mass does not change and justifies this mainly by appealing to
conservation of mass, which is not the correct reasoning: total mass staying the same does not
imply the centre of mass is unchanged. In reality, the ice's mass moves from a higher position
(block) to a thin layer at the bottom, so the system's centre of mass shifts downward. The response
also ignores the requested discussion of how the relative masses of ice and tray affect the
magnitude of this shift. To improve, the student should: (1) state that the centre of mass moves
downward because the ice's mass is redistributed closer to the tray's base; (2) explain that the
tray's own centre of mass is unchanged; and (3) discuss that the shift is larger when the ice mass
is comparable to or larger than the tray's mass, and smaller (almost negligible) when the tray is
much heavier than the ice.

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/playground/subjective/9gkqeb7ttnk6
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WHERE THESE QUESTIONS CAME FROM (sources)

- **Subject assignment pages / My Assignments + Assignment Contests API** — 1 assignments, 1 questions; full statement captured from each question's playground page (read-only visit) — 1/1 texts present.
- **Lecture pages** (Course Content sidebar list) — 0 of 26 lecture windows carry content: post-class question text below the video (`Read More` expanded) and in-class quiz questions via read-only views only (Revise / View Solutions / Next-Previous walk — never Attempt/Start/Submit).
- **Assessments API** — 0 quiz schedules (title, window, XP earned). No question text in the API (hidden until test start).

NOT FETCHED — open these yourself

(These sit behind `Attempt Now` / `Start Test` buttons. I never click those — they consume quiz attempts. Everything else above is complete.)

Quiz Contests tab (separate MCQ tests)

- Quiz contest questions only appear after `Start Test` (graded, consumes an attempt): open the Quiz Contests tab manually if you want these: PSP Contest 1, Wifi Demo, Demo_Contest, Mock Test
- What IS captured: all assignment questions (1), all in-class quiz questions shown above, all contest ASSIGNMENT texts

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TOTALS: 1 assignment questions (0 solved), 0 quizzes, 0 questions without captured text, 0 lecture windows with content